



Retail Sales Data Analytics Report

1. Introduction

The objective of this project is to perform **Exploratory Data Analysis (EDA)** on a retail sales dataset to understand sales trends, customer purchasing behavior, and key revenue drivers. The analysis aims to derive **actionable business insights** using descriptive statistics, time-series analysis, correlation analysis, and visualizations.

2. Dataset Overview

The dataset consists of **1000 retail transactions** with the following key attributes: - Transaction ID - Date - Customer ID - Gender - Age - Product Category - Quantity - Price per Unit - Total Amount

The data represents individual purchase transactions across multiple product categories and customers.

3. Data Loading and Cleaning

- The dataset was loaded using the Pandas library.
- Data types were validated using `df.info()`.
- The `Date` column was converted from object type to `datetime` format to support time-based analysis.
- No missing values or duplicate records were found, ensuring data quality.

Outcome: The dataset was clean and ready for analysis.

4. Descriptive Statistics

Basic descriptive statistics such as **mean, median, standard deviation, minimum, maximum, and quartiles** were calculated for numerical variables.

Key Observations:

- The **median transaction value** is relatively low, indicating that most purchases are low-value.
 - The **mean is higher than the median**, suggesting a right-skewed distribution driven by fewer high-value purchases.
 - Quartile analysis of price per unit shows a wide range, indicating the presence of **budget, mid-range, and premium-priced products**.
-

5. Time Series Analysis

Monthly sales trends were analyzed using the Date column.

Key Observations:

- Sales show **monthly fluctuations**, indicating seasonality.
- Certain months contribute significantly more to total revenue, suggesting periods of high demand.

Business Value: Helps in planning promotions, inventory management, and staffing.

6. Customer and Product Analysis

Customer Behavior

- Purchase behavior varies significantly across customers.
- Customers buying the same quantity often spend different total amounts due to variation in product prices.

Product Category Analysis

- Some product categories contribute disproportionately to total sales.
 - High-priced products, even with lower quantities, significantly impact revenue.
-

7. Purchase Behaviour Analysis

Scatter plot analysis between **Quantity and Total Amount** revealed: - A positive but moderate relationship. - Customers purchasing more items generally spend more, but product pricing plays a larger role.

This highlights that **price per unit is a stronger driver of revenue than quantity alone**.

8. Outlier Detection

Outliers were analyzed using boxplots and the IQR method.

Findings:

- No extreme outliers were detected beyond the whiskers.
 - High-value transactions fall within the expected range and represent **valid premium or bulk purchases**, not anomalies.
-

9. Correlation Analysis

Correlation analysis was performed among numerical variables.

Key Results:

- **Price per Unit and Total Amount** show a **strong positive correlation**, indicating pricing as the primary revenue driver.
- **Quantity and Total Amount** show a **moderate correlation**, as price variability affects total spending.
- **Age has negligible correlation** with spending behavior.

Conclusion: Revenue is primarily driven by product pricing rather than customer demographics.

10. Key Insights Summary

- Most transactions are low-value, but a small number of high-value purchases significantly impact revenue.
 - Product pricing has a stronger influence on total sales than quantity.
 - Customer age does not significantly affect purchasing behavior.
 - Sales exhibit seasonal patterns.
 - Premium-priced products contribute heavily to overall revenue.
-

11. Business Recommendations

- Focus marketing efforts on **high-performing product categories**.
 - Promote **premium-priced products** to maximize revenue.
 - Introduce **bundled offers or discounts** for budget buyers to increase volume.
 - Plan targeted campaigns during **peak sales months**.
 - Optimize inventory based on seasonal demand trends.
-

12. Conclusion

This retail sales data analytics project provided a comprehensive understanding of customer behavior, sales trends, and revenue drivers. Through effective EDA and visualization, the analysis delivered meaningful insights and actionable recommendations to support data-driven business decisions.

End of Report